

# PROJECT PROPOSAL

RareSense.AI — LLM-Powered Rare Disease Detection from Unstructured Clinical Notes

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**Syllabus Equivalent:** Project #7 (CareerPulse)

**Domain:** Health Data Science (HDS)

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## 1. Problem Statement

Rare diseases affect over **300 million people worldwide**, yet the average rare disease takes **7.6 years to diagnose**. Patients visit an average of 7 specialists before receiving a correct diagnosis. The core issue is that symptoms are scattered across years of fragmented clinical notes written by different doctors, and no single physician sees the complete picture. Traditional keyword-based search systems fail because doctors use varied terminology (e.g., 'NIDDM' vs 'Type 2 Diabetes' vs 'non-insulin dependent diabetes').

## 2. Proposed Solution: RareSense.AI

**RareSense.AI** is an intelligent diagnostic support system that reads unstructured clinical notes, extracts medical entities (symptoms, diagnoses, medications, lab values) using Clinical NLP, builds a patient health timeline, and cross-references it against rare disease databases (OMIM / Orphanet) to flag potential undiagnosed rare conditions that may have been missed.

## 3. System Architecture

- **Data Source:** MIMIC-IV — a publicly available dataset of real ICU patient records from Beth Israel Deaconess Medical Center (a Harvard-affiliated teaching hospital). Contains 40,000+ patient stays with clinical notes, lab results, medications, and ICD codes.
- **Clinical NLP Engine:** Uses pre-trained BioBERT and medSpaCy for Named Entity Recognition (NER) to extract symptoms, diagnoses, medications, and lab findings from free-text doctor notes.
- **Patient Timeline Builder:** Constructs a chronological health timeline per patient in MongoDB, linking extracted entities across multiple clinical encounters over time.
- **Rare Disease Matching Engine:** Generates phenotype embeddings from the patient timeline using MongoDB Vector Search and calculates semantic similarity against rare disease symptom profiles from the Orphanet and OMIM databases.
- **Clinician Dashboard:** A React.js interface displaying flagged rare disease candidates with confidence scores, matched symptom evidence, and a visual patient health timeline.

## 4. Technical Equivalence to CareerPulse (Project #7)

The **database schema, algorithmic complexity, and coding deliverables remain 100% equivalent** to the originally assigned CareerPulse project:

| Core Module | Original: CareerPulse | Proposed: RareSense.AI |
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|                     |                                                                                           |                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Data Ingestion      | Ingests user resumes/portfolios with flexible schemas for varying skills and experiences. | Ingests clinical notes from MIMIC-IV with flexible schemas for varying medical specialties and record formats.                |
| Search & Matching   | Recommends job vacancies by matching candidate skill profiles against job criteria.       | Matches patient symptom profiles against rare disease criteria using MongoDB Vector Search and clinical embeddings (BioBERT). |
| Activity Tracking   | Tracks course completions, application stages, and user activity over time.               | Tracks patient symptoms, medications, and lab results chronologically to build a health timeline.                             |
| Analytics Dashboard | Visualizes hiring trends, salary data, and skill progression graphs.                      | Visualizes rare disease match scores, symptom evidence, and patient health timelines.                                         |

## 5. Health Data Science Significance

Rare disease diagnostics represents one of the most critical unmet needs in modern clinical informatics. By applying clinical NLP and semantic vector matching to real-world EHR data, this project demonstrates advanced health data engineering capabilities and addresses a problem that directly impacts patient lives. The use of MIMIC-IV (a publicly available clinical dataset from a leading academic medical center) ensures reproducibility and academic rigor.

## 6. Technology Stack

- **Database:** MongoDB (Document Store + Vector Search + Time-Series)
- **NLP:** BioBERT, medSpaCy, scispaCy (pre-trained clinical models)
- **Backend:** Python (FastAPI)
- **Frontend:** React.js
- **Data:** MIMIC-IV, Orphanet API, OMIM